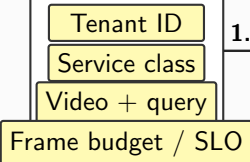
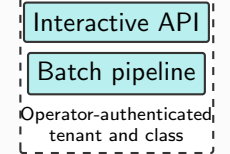


Tenant applications

Multimodal request

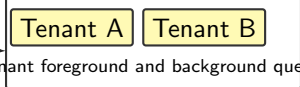


Execution context



1. Submit

Admission and queue manager

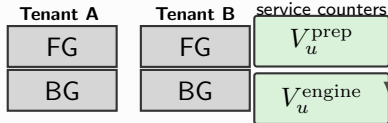


Per-tenant foreground and background queues

2. Enqueue

Hierarchical scheduler

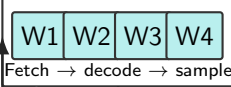
1. Select tenant by normalized virtual service
2. Select foreground before background within tenant



Separate stage service counters

Cross-stage multimodal serving system

Media preparation

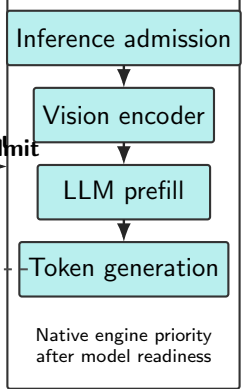


Fetch → decode → sample

Bounded model-ready queue



VLM inference tier



change pre service

4. Admit

charge engine service

Work-conserving borrowing · background aging · non-preemptive active decode